

Flood Monitoring System – Data Dictionary

Table of Contents

Table: ALERT.....	2
Table: ALERT_RULE	3
Table: STATION.....	5
Table: SENSOR.....	6
Table: SENSOR_TYPE.....	7
Table: MEASUREMENT	7
Table: FLOOD_EVENT.....	8
Table: EVENT_IMPACT.....	10
Table: MODEL_RUN	11
Table: MODEL_INPUT	12
Table: MODEL_OUTPUT.....	13
Table: DATA_SOURCE	15
Table: INGEST_JOB.....	16
Table: INGEST_ERROR	17
Table: NOTIFICATION	18
Table: ROLE	19
Table: USER.....	20
Table: USER_AREA_SUBSCRIPTION	21
Table: USER_SESSION	22
Table: AUDIT_LOG.....	23
Table: GEOGRAPHICAL_AREA.....	24

Table: ALERT

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
ALERTID	bigint	No	PK	—	Primary Key	Unique identifier for each alert event.	Auto-generated; immutable once created.
ALERTRULEID	int	No	FK	—	FK → ALERT_RULE (AlertRuleID, StationID)	References the alert rule that triggered the alert.	Must match existing composite rule with same StationID.
STATIONID	int	No	FK	—	FK → ALERT_RULE (AlertRuleID, StationID)	Monitoring station where alert originated.	Must correspond to same station defined in alert rule.
USERID	int	No	FK	—	FK → USER(UserID)	User associated with the alert.	Must reference an existing active user.
TRIGGEREDAT	datetime2	No	—	sysutcdatetime()	Default constraint	UTC timestamp when alert was triggered.	System-generated at insert time.
SEVERITY	nvarchar(20)	No	—	—	CHECK (LOW, MEDIUM, HIGH, CRITICAL)	Severity classification of alert.	Must match predefined domain values.
STATUS	nvarchar(30)	No	—	—	CHECK (TRIGGERED, ACKNOWLEDGED, RESOLVED, FALSE_POSITIVE)	Current lifecycle state of alert.	Controls workflow progression.
SUMMARY	nvarchar(1000)	Yes	—	—	—	Human-readable short description.	Optional but recommended for

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
							operational clarity.
DETAILSJSON	nvarchar(MAX)	Yes	—	—	CHECK ISJSON()	Structured metadata for alert context.	Must be valid JSON if provided.
ACKNOWLEDGEDAT	datetime2	Yes	—	—	—	Timestamp when alert was acknowledged.	Must be ≥ TriggeredAt if populated.

Table: ALERT_RULE

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
ALERTRULEID	int	No	PK	—	Primary Key	Unique identifier for an alert rule.	System-generated; uniquely identifies rule logic.
STATIONID	int	No	FK	—	FK → STATION(StationID)	Station to which the rule applies.	Rule is scoped to a single station.
AREAID	int	No	FK	—	FK → GEOGRAPHIC_AREA(AreaID)	Geographic area associated with the rule.	Must correspond to the station's area.
RULENAME	nvarchar(120)	No	—	—	—	Human-readable name of the alert rule.	Should be unique per station (logical uniqueness).

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
METRIC	nvarchar(80)	No	—	—	—	Metric being evaluated (e.g., WaterLevel).	Must correspond to a measurable sensor metric.
OPERATOR	nvarchar(10)	No	—	—	CHECK (!=, =, <, <=, >, >=)	Comparison operator used in threshold logic.	Defines logical evaluation behavior.
THRESHOLDVALUE	decimal(18,6)	No	—	—	—	Threshold value used for rule evaluation.	Precision allows scientific-grade measurement comparisons.
WINDOWMINUTES	int	No	—	—	CHECK (> 0)	Rolling evaluation window in minutes.	Must be positive; defines aggregation period.
SEVERITY	nvarchar(20)	No	—	—	CHECK (LOW, MEDIUM, HIGH, CRITICAL)	Severity level assigned when rule triggers.	Propagated to ALERT table on trigger.
ISENABLED	bit	No	—	1	Default constraint	Indicates whether rule is active.	Disabled rules cannot generate alerts.
CREATEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp rule was created.	System-generated; immutable.

Table: STATION

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
STATIONID	int	No	PK	—	Primary Key	Unique identifier for each monitoring station.	System-generated; immutable.
AREAID	int	No	FK	—	FK → GEOGRAPHIC_AREA(AreaID)	Geographic area where the station is located.	Must reference a valid geographic area.
STATIONCODE	nvarchar(50)	No	UQ	—	UNIQUE (StationCode)	Short operational code used to identify the station.	Must be globally unique.
STATIONNAME	nvarchar(120)	No	—	—	—	Full descriptive name of the monitoring station.	Human-readable identifier.
LATITUDE	decimal(9,6)	No	—	—	CHECK (-90 ≤ Latitude ≤ 90)	Latitude coordinate in decimal degrees.	Required for spatial positioning.
LONGITUDE	decimal(9,6)	No	—	—	CHECK (-180 ≤ Longitude ≤ 180)	Longitude coordinate in decimal degrees.	Required for spatial positioning.
ELEVATIONM	decimal(10,2)	Yes	—	—	CHECK (ElevationM ≥ 0)	Elevation above sea level in meters.	Optional; must be non-negative if provided.
STATUS	nvarchar(30)	No	—	N'ACTIVE'	CHECK (ACTIVE, INACTIVE, MAINTENANCE)	Operational status of station.	Controls whether station can generate measurements.
INSTALLEDAT	datetime2	Yes	—	—	—	Date the station was installed.	Must be ≤ current UTC date if provided.

Table: SENSOR

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
SENSORID	int	No	PK	—	Primary Key	Unique identifier for each sensor device.	System-generated; immutable.
STATIONID	int	No	FK	—	FK → STATION(StationID)	Station where the sensor is installed.	Sensor must belong to exactly one station.
SENSORTYPEID	int	No	FK	—	FK → SENSOR_TYPE(SensorTypeID)	Type/category of sensor (e.g., WaterLevel).	Must reference a valid sensor type definition.
SERIALNUMBER	nvarchar(80)	Yes	—	—	—	Manufacturer or internal serial number.	Optional; should be unique in real deployments (recommended).
STATUS	nvarchar(30)	No	—	N'ACTIVE'	CHECK (ACTIVE, INACTIVE, MAINTENANCE, RETIRED)	Operational state of the sensor.	Inactive/retired sensors should not produce valid measurements.
INSTALLEDAT	datetime2	Yes	—	—	—	Timestamp when sensor was installed.	Must be ≤ current UTC date if provided.
LASTCALIBRATIONAT	datetime2	Yes	—	—	—	Timestamp of last calibration.	Must be ≥ InstalledAt if both are present.

Table: SENSOR_TYPE

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
SENSORTYPEID	int	No	PK	—	Primary Key	Unique identifier for each sensor type.	System-generated; immutable.
TYPENAME	nvarchar(80)	No	UQ	—	UNIQUE (TypeName)	Name of the sensor type (e.g., WaterLevel, RainfallRate, FlowRate).	Must be globally unique.
UNIT	nvarchar(32)	Yes	—	—	—	Measurement unit associated with the sensor type (e.g., meters, mm/hr, m ³ /s).	Should follow standardized unit conventions.
DESCRIPTION	nvarchar(255)	Yes	—	—	—	Human-readable explanation of what the sensor type measures.	Optional documentation field.

Table: MEASUREMENT

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
MEASUREMENTID	bigint	No	PK	—	Primary Key	Unique identifier for each recorded measurement.	System-generated; immutable.
STATIONID	int	No	FK	—	Part of composite FK → SENSOR(SensorID, StationID)	Station where the measurement was recorded.	Must match the station of the referenced sensor.
SENSORID	int	No	FK	—	Part of composite FK → SENSOR(SensorID, StationID)	Sensor that produced the measurement.	Must belong to the specified StationID.

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
OBSERVEDAT	datetime2	No	—	—	—	UTC timestamp when measurement was observed.	Cannot be in the future (business rule).
VALUE	decimal(18,6)	No	—	—	—	Numeric measurement value.	Precision supports hydrological and scientific data.
UNIT	nvarchar(32)	Yes	—	—	—	Unit of the measurement.	Should align with SENSOR_TYPE.Unit; optional redundancy for traceability.
QUALITYFLAG	nvarchar(20)	Yes	—	—	CHECK (VALID, ESTIMATED, MISSING, OUTLIER)	Data quality classification.	Supports downstream filtering and model validation.
INGESTEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when measurement was stored in the system.	Must be ≥ ObservedAt.

Table: FLOOD_EVENT

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
FLOODEVENTID	bigint	No	PK	—	Primary Key	Unique identifier for a detected flood event.	System-generated; immutable.

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
MODELRUNID	bigint	No	FK	—	FK → MODEL_RUN(ModelRunID) (onDelete: CASCADE)	Model run that produced the detection.	A flood event must be attributable to exactly one model run.
AREAID	int	No	FK	—	FK → GEOGRAPHIC_AREA(AreaID)	Geographic area where the event was detected.	Event must belong to a valid area.
DETECTEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when the event was detected in the system.	System-generated; should not be in the future.
EVENTSTART	datetime2	Yes	—	—	—	Estimated start time of the flood event.	If present, must be ≤ EventEnd.
EVENTEND	datetime2	Yes	—	—	—	Estimated end time of the flood event.	If present, must be ≥ EventStart.
CONFIDENCE	nvarchar(30)	Yes	—	—	—	Model confidence level (e.g., LOW/MEDIUM/HIGH or probabilistic label).	Should be standardized across model versions.
SUMMARY	nvarchar(1000)	Yes	—	—	—	Human-readable summary of detection.	Optional but recommended for reporting.

Table: EVENT_IMPACT

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
IMPACTID	bigint	No	PK	—	Primary Key	Unique identifier for a single impact record.	System-generated; immutable.
FLOOD_EVENTID	bigint	No	FK	—	FK → FLOOD_EVENT(FloodEventID) (onDelete: CASCADE)	Flood event this impact record belongs to.	Must reference an existing flood event.
IMPACTTYPE	nvarchar(50)	No	—	—	—	Category of impact (e.g., Inundation, Infrastructure, Population, Agriculture).	Should be standardized for reporting consistency.
SEVERITY	nvarchar(20)	Yes	—	—	—	Severity label for this specific impact.	If used, should align with system severity domain.
ESTIMATEDAREASQKM	decimal(12,2)	Yes	—	—	CHECK (>= 0)	Estimated affected area in square kilometers.	Must be non-negative if provided.
ESTIMATEDPEOPLEAFFECTED	int	Yes	—	—	CHECK (>= 0)	Estimated number of people	Must be non-negative if provided.

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
						affected.	
NOTES	nvarchar(1000)	Yes	—	—	—	Free-text qualitative notes about the impact.	Optional; useful for analyst interpretation and auditing.

Table: MODEL_RUN

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
MODELRUNID	bigint	No	PK	—	Primary Key	Unique identifier for a model execution instance.	System-generated; immutable.
MODELNAME	nvarchar(120)	No	—	—	—	Name of the analytical model executed.	Should be consistent with deployment registry naming.
MODELVERSION	nvarchar(50)	No	—	—	—	Version identifier of the model used.	Must reflect exact deployed version for reproducibility.
RUNSTARTEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when execution began.	System-generated at run start.
RUNCOMPLETEDAT	datetime2	Yes	—	—	—	Timestamp when execution completed.	Must be $\geq$ RunStartedAt if present.

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
STATUS	nvarchar(30)	No	—	—	CHECK (QUEUED, RUNNING, COMPLETED, FAILED)	Execution lifecycle state.	Controls orchestration and downstream processing.
PARAMETERSJSON	nvarchar(MAX)	Yes	—	—	CHECK ISJSON()	Serialized configuration parameters for the run.	Must be valid JSON if provided.
NOTES	nvarchar(1000)	Yes	—	—	—	Analyst notes or execution comments.	Optional documentation field.

Table: MODEL_INPUT

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
MODELINPUTID	bigint	No	PK	—	Primary Key	Unique identifier for a model input record.	System-generated; immutable.
MODELRUNID	bigint	No	FK	—	FK → MODEL_RUN(ModelRunID) (onDelete: CASCADE)	The model run this input belongs to.	A model input cannot exist without its model run.
INPUTTYPE	nvarchar(50)	No	—	—	—	Category/type of input (e.g., Measurements, SatelliteRaster, WeatherForecast,	Should be standardized across runs for reporting.

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
						DEM).	
SOURCEREF	nvarchar(500)	Yes	—	—	—	Reference/location of the input source (URI/path/object key).	Should be resolvable to retrieve the exact dataset used.
TIMESTART	datetime2	Yes	—	—	—	Start of time range covered by the input.	Must be ≤ TimeEnd if both are present.
TIMEEND	datetime2	Yes	—	—	—	End of time range covered by the input.	Must be ≥ TimeStart if both are present.
INPUTMETADATAJSON	nvarchar(MAX)	Yes	—	—	CHECK ISJSON()	Structured metadata describing the input (schema, bands, preprocessing, etc.).	Must be valid JSON if provided.

Table: MODEL_OUTPUT

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
MODELOUTPUTID	bigint	No	PK	—	Primary Key	Unique identifier for a model output record.	System-generated; immutable.

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
MODELRUNID	bigint	No	FK	—	FK → MODEL_RUN(ModelRunID) (onDelete: CASCADE)	The model run that produced this output.	Cannot exist without parent model run.
OUTPUTTYPE	nvarchar(50)	No	—	—	—	Type/category of output (e.g., FloodExtentRaster, RiskScoreGrid, ProbabilityMap, ClassificationResult).	Should be standardized for reporting and retrieval.
OUTPUTREF	nvarchar(500)	Yes	—	—	—	Reference or storage location of output artifact (URI/path/object key).	Should allow retrieval of exact artifact used in event derivation.
OUTPUTMETADATAJSON	nvarchar(MAX)	Yes	—	—	CHECK ISJSON()	Structured metadata describing output properties (resolution, CRS, bands, thresholds).	Must be valid JSON if populated.
PRODUCEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when output artifact was generated.	Must be ≥ ModelRun.RunStartedAt.

Table: DATA_SOURCE

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
DATASOURCEID	int	No	PK	—	Primary Key	Unique identifier for a registered data source.	System-generated; immutable.
SOURCENAME	nvarchar(120)	No	—	—	—	Human-readable name of the data source.	Should be unique by policy (recommended).
SOURCETYPE	nvarchar(50)	No	—	—	—	Category of source (e.g., API, FILE, SATELLITE, STREAM, DATABASE).	Should be standardized across the system.
PROVIDER	nvarchar(120)	Yes	—	—	—	Organization/vendor providing the data.	Optional; supports provenance and reporting.
ENDPOINT	nvarchar(500)	Yes	—	—	—	Connection endpoint or resource locator (URL/path/bucket).	Must be valid and reachable for active sources.
AUTHMETHOD	nvarchar(50)	Yes	—	—	—	Authentication method (e.g., NONE, API_KEY, OAUTH, IAM_ROLE).	Must align with operational integration pattern.
ISACTIVE	bit	No	—	1	Default constraint	Indicates whether source is enabled for ingestion.	Inactive sources should not schedule new ingest jobs.
CREATEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when the source was registered.	System-generated; immutable.

Table: INGEST_JOB

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
INGESTJOBID	int	No	PK	—	Primary Key	Unique identifier for an ingestion job execution.	System-generated; immutable.
DATASOURCEID	int	No	FK	—	FK → DATA_SOURCE(DataSourceID)	Data source from which ingestion was performed.	Must reference an active data source.
JOBTYPE	nvarchar(50)	No	—	—	—	Type of ingestion (e.g., FULL_LOAD, INCREMENTAL, REALTIME, BACKFILL).	Should follow standardized operational taxonomy.
STATUS	nvarchar(30)	No	—	—	CHECK (QUEUED, RUNNING, COMPLETED, FAILED)	Current lifecycle state of ingestion job.	Controls operational monitoring and retry logic.
STARTEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when ingestion began.	System-generated at job start.
COMPLETEDAT	datetime2	Yes	—	—	—	Timestamp when ingestion finished.	Must be ≥ StartedAt if populated.
PARAMETERSJSON	nvarchar(MAX)	Yes	—	—	CHECK ISJSON()	Configuration parameters used for ingestion (filters,	Must be valid JSON if

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
						offsets, date ranges).	provided.
OUTPUTLOCATION	nvarchar(500)	Yes	—	—	—	Storage location of processed/ingested data.	Should allow retrieval of ingest artifacts.

Table: INGEST_ERROR

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
INGESTERRORID	int	No	PK	—	Primary Key	Unique identifier for an ingestion error record.	System-generated; immutable.
INGESTJOBID	int	No	FK	—	FK → INGEST_JOB(IngestJobID) (onDelete: CASCADE)	Ingestion job during which the error occurred.	Cannot exist without a parent ingest job.
ERRORCODE	nvarchar(50)	Yes	—	—	—	Optional short code for error categorization.	Should be standardized where possible.
ERRORMESSAGE	nvarchar(4000)	No	—	—	—	Full error message describing the failure.	Required for troubleshooting and audit.
OCCURREDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when the error occurred.	System-generated at insert time.
CONTEXTJSON	nvarchar(MAX)	Yes	—	—	CHECK ISJSON()	Structured diagnostic context	Must be valid

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
						(payload snippet, stack trace, request metadata).	JSON if provided.

Table: NOTIFICATION

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
NOTIFICATIONID	bigint	No	PK	—	Primary Key	Unique identifier for a notification attempt.	System-generated; immutable.
ALERTID	bigint	No	FK	—	FK → ALERT(AlertID) (OnDelete: CASCADE)	Alert that triggered this notification.	Cannot exist without a valid alert.
USERID	int	No	FK	—	FK → USER(UserID)	Recipient of the notification.	Must reference an existing user.
CHANNEL	nvarchar(20)	No	—	—	CHECK (EMAIL, SMS, PUSH, WEBHOOK)	Delivery channel used to send notification.	Must match supported communication mechanisms.
DESTINATION	nvarchar(256)	No	—	—	—	Target address (email address, phone number, webhook URL, etc.).	Must be valid for selected channel.
STATUS	nvarchar(30)	No	—	—	CHECK (QUEUED, SENT, FAILED)	Current delivery state of notification.	Reflects message delivery lifecycle.

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
QUEUEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when notification was queued.	System-generated at creation.
SENTAT	datetime2	Yes	—	—	—	Timestamp when notification was sent.	Must be ≥ QueuedAt if populated.
PROVIDERMESAGEID	nvarchar(128)	Yes	—	—	—	External provider message identifier.	Used for reconciliation with messaging provider logs.

Table: ROLE

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
ROLEID	int	No	PK	—	Primary Key	Unique identifier for a role.	System-generated; immutable.
ROLENAME	nvarchar(50)	No	UQ	—	UNIQUE (RoleName)	Name of the role (e.g., ADMIN, ANALYST, OPERATOR, VIEWER).	Must be globally unique.
ROLEDESCRIPTION	nvarchar(255)	Yes	—	—	—	Human-readable explanation of permissions associated with role.	Optional documentation field.
CREATEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when role was created.	System-generated; immutable.

Table: USER

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
USERID	int	No	PK	—	Primary Key	Unique identifier for a system user.	System-generated; immutable.
ROLEID	int	No	FK	—	FK → ROLE(RoleID)	Role assigned to the user.	Must reference a valid role.
EMAIL	nvarchar(320)	No	UQ	—	UNIQUE (Email)	User's email address used for login and notifications.	Must be globally unique.
DISPLAYNAME	nvarchar(120)	No	—	—	—	Human-readable name shown in system interfaces.	Required for operational clarity.
PASSWORDHASH	varbinary(256)	No	—	—	—	Hashed password value.	Must never store plaintext credentials.
ISACTIVE	bit	No	—	1	Default constraint	Indicates whether user account is active.	Inactive users cannot authenticate.
CREATEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when user account was created.	System-generated; immutable.
LASTLOGINAT	datetime2	Yes	—	—	—	Timestamp of most recent successful login.	Updated upon authentication.

Table: USER_AREA_SUBSCRIPTION

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
SUBSCRIPTIONID	int	No	PK	—	Primary Key	Unique identifier for a subscription record.	System-generated; immutable.
USERID	int	No	FK	—	FK → USER(UserID) (OnDelete: CASCADE)	User who owns this subscription.	Cannot exist without a valid user.
AREAID	int	No	FK	—	FK → GEOGRAPHIC_AREA(AreaID)	Geographic area being subscribed to.	Must reference an existing area.
PREFERENCEJSON	nvarchar(MAX)	Yes	—	—	CHECK ISJSON()	Subscription configuration (channels, thresholds, quiet hours, etc.).	Must be valid JSON if provided.
ISACTIVE	bit	No	—	1	Default constraint	Indicates whether subscription is active.	Inactive subscriptions do not generate notifications.
CREATEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when subscription was created.	System-generated; immutable.

Table: USER_SESSION

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
SESSIONID	int	No	PK	—	Primary Key	Unique identifier for a user session record.	System-generated; immutable.
USERID	int	No	FK	—	FK → USER(UserID) (onDelete: CASCADE)	User who owns this session.	Cannot exist without a valid user.
REFRESHTOKENHASH	varbinary(256)	No	—	—	—	Hashed refresh token associated with the session.	Never store raw refresh tokens. Must be cryptographically derived.
ISSUEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when session was issued.	System-generated at creation.
EXPIRESAT	datetime2	No	—	—	—	Timestamp when session expires.	Must be > IssuedAt.
REVOKEDAT	datetime2	Yes	—	—	—	Timestamp when session was revoked.	If populated, must be ≥ IssuedAt and ≤ ExpiresAt.
CLIENTIP	nvarchar(45)	Yes	—	—	—	IP address observed at session issuance/use (IPv4/IPv6).	Optional but recommended for security auditing.
USERAGENT	nvarchar(256)	Yes	—	—	—	Client user-agent string.	Optional; supports fraud detection and troubleshooting.

Table: AUDIT_LOG

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
AUDITLOGID	int	No	PK	—	Primary Key	Unique identifier for an audit log record.	System-generated; immutable.
USERID	int	No	FK	—	FK → USER(UserID)	User who performed the action.	Must reference a valid user.
ACTIONTYPE	nvarchar(50)	No	—	—	—	Type of action performed (e.g., CREATE, UPDATE, DELETE, LOGIN, LOGOUT, ACK_ALERT).	Should be standardized for consistent reporting.
ENTITYTYPE	nvarchar(50)	No	—	—	—	Entity/table affected (e.g., ALERT, STATION, USER, MODEL_RUN).	Should align with system entity naming.
ENTITYID	nvarchar(64)	No	—	—	—	Identifier of the entity instance affected.	Stored as text to support mixed key types across entities.
METADATAJSON	nvarchar(MAX)	Yes	—	—	CHECK ISJSON()	Structured audit metadata (fields changed, request ID, before/after snapshots).	Must be valid JSON if provided.
OCCURREDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when the action occurred.	System-generated at insert time.
SOURCEIP	nvarchar(45)	Yes	—	—	—	IP address from which action originated.	Optional but recommended; supports investigations.

Table: GEOGRAPHIC_AREA

COLUMN NAME	DATA TYPE	NULLABLE	KEY TYPE	DEFAULT	CONSTRAINTS	DESCRIPTION	BUSINESS RULES
AREAID	int	No	PK	—	Primary Key	Unique identifier for a geographic area.	System-generated; immutable.
AREANAME	nvarchar(120)	No	—	—	—	Human-readable name of the area (e.g., “Dartmouth Test Area”).	Should be unique by policy (recommended).
AREATYPE	nvarchar(50)	No	—	—	—	Classification of area (e.g., CITY, REGION, BASIN, WATERSHED, ADMIN).	Should be standardized for consistent use.
GEOMETRYWKT	nvarchar(MAX)	Yes	—	—	—	Area geometry in Well-Known Text (WKT) format.	Optional; must be valid WKT by policy if stored.
MINLAT	decimal(9,6)	Yes	—	—	CHECK (-90 ≤ MinLat ≤ 90)	Minimum latitude of area bounding box.	If bounding box fields are used, all should be present.
MINLON	decimal(9,6)	Yes	—	—	CHECK (-180 ≤ MinLon ≤ 180)	Minimum longitude of area bounding box.	Must be valid geographic longitude.
MAXLAT	decimal(9,6)	Yes	—	—	CHECK (-90 ≤ MaxLat ≤ 90)	Maximum latitude of area bounding box.	Must be ≥ MinLat if both present.
MAXLON	decimal(9,6)	Yes	—	—	CHECK (-180 ≤ MaxLon ≤ 180)	Maximum longitude of area bounding box.	Must be ≥ MinLon if both present.
CREATEDAT	datetime2	No	—	sysutcdatetime()	Default constraint	Timestamp when the area record was created.	System-generated; immutable.